

# NetworkFP Analytics Instrumentation Agent

*An AI-powered workflow that automatically crawls a product, generates structured Mixpanel event taxonomy, and delivers developer-ready output*

Built in: 1 day | Cost: ₹0/month | Code written: 0 lines

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## 1. Project Overview

This project was built to solve a real analytics problem at NetworkFP - a financial learning platform for wealth managers and financial advisors. The platform had Mixpanel integrated but zero events configured, meaning there was no visibility into what users were actually doing.

Instead of manually auditing every page and writing events by hand - a process that would take days - an AI agent was built that does this automatically every day.

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## 2. Problem Statement

### The Situation

NetworkFP ([learn.networkfp.com](https://learn.networkfp.com)) is a learning platform for financial professionals. It contains courses, sessions, toolbox resources, exchange forums, quizzes, and client connect materials.

Mixpanel was integrated into the platform but no tracking events were set up. This meant:

- No data on which pages users visited
- No data on which courses or sessions were most popular
- No data on where users dropped off or lost interest
- No data on which resources were downloaded
- No way to make informed product decisions

### The Manual Alternative

The traditional approach would have been to manually visit every page, document every possible user action, write event names and properties following naming conventions, prioritise which events matter most, and hand the list to a developer. For a platform with 50+ pages, this would take 20-30 days if 1 hour is dedicated daily.

### The Key Insight

*If an AI can read a page and understand what actions a user can take on it, it can write the events automatically - and do it every day for every page without any human involvement.*

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### 3. The Solution

#### What Was Built

A fully automated n8n workflow that runs every morning at 11am and:

- Crawls learn.networkfp.com automatically using Firecrawl
- Sends each page's content to GPT-4o-mini with a custom analytics prompt
- AI analyses the page and writes all relevant Mixpanel events
- Every event includes: name, properties, priority (High/Medium/Low), and reason
- All events are automatically appended to a Google Sheet
- Developer picks up the sheet and implements the events directly

#### The Stack

| Tool               | Purpose          | Cost      | Free Tier    |
|--------------------|------------------|-----------|--------------|
| n8n                | Workflow builder | Free      | Yes          |
| Firecrawl          | Website crawler  | Free      | 500 pages/mo |
| OpenAI GPT-4o-mini | AI reasoning     | ~₹1/month | Free credits |
| Google Sheets      | Output storage   | Free      | Yes          |

#### Why This Is An Agent, Not Just Automation

A standard automation does the same thing every time with no reasoning. What makes this an agent is the AI reasoning step in the middle. The AI:

- Reads a page it has never seen before
- Understands what kind of page it is (course listing vs forum vs quiz)
- Figures out what user actions are possible on that specific page
- Applies product analytics best practices
- Prioritises events by business importance
- Writes different events for different pages based on context

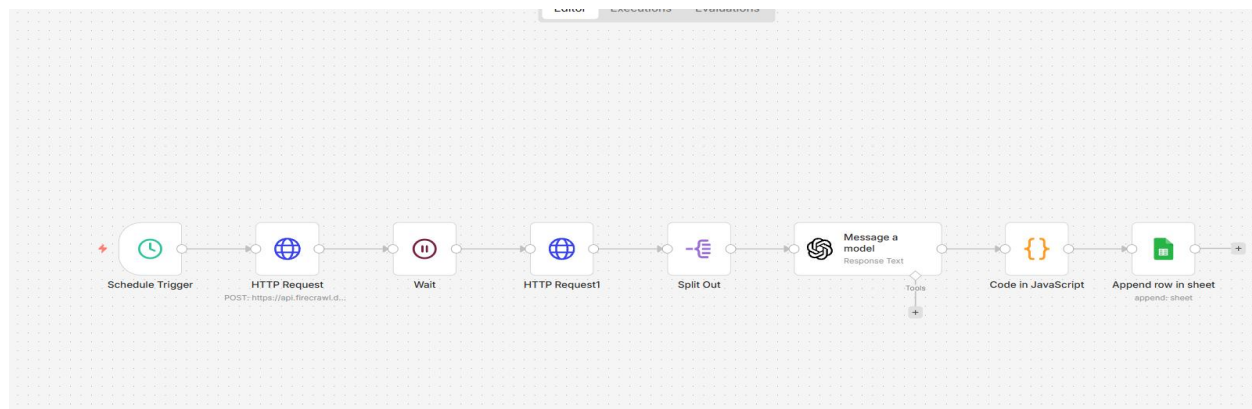
No two pages produce the same output. The AI makes contextual decisions - that is what makes it agentic.

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### 4. How It Works - Node by Node

The workflow consists of 8 nodes in n8n, each doing one specific job:

| # | Node                       | What it does                                                  | Tool used          |
|---|----------------------------|---------------------------------------------------------------|--------------------|
| 1 | <b>Schedule Trigger</b>    | Runs workflow every day at 11am                               | Clock / Timer      |
| 2 | <b>HTTP Request (POST)</b> | Tells Firecrawl to start crawling the website; limit 10 pages | Firecrawl API      |
| 3 | <b>Wait Node</b>           | Pauses 60 seconds for crawl to complete                       | n8n built-in       |
| 4 | <b>HTTP Request (GET)</b>  | Fetches crawled page content and URLs                         | Firecrawl API      |
| 5 | <b>Split Out</b>           | Breaks 10 pages into individual items                         | n8n built-in       |
| 6 | <b>Message a Model</b>     | AI analyses each page, writes Mixpanel events                 | OpenAI GPT-4o-mini |
| 7 | <b>Code (JavaScript)</b>   | Parses AI JSON output into clean data rows                    | n8n built-in       |
| 8 | <b>Google Sheets</b>       | Appends each event as a new row in the sheet                  | Google Sheets API  |



## The Flow Visualised

```

Schedule Trigger (11am daily)
↓
HTTP Request → POST to Firecrawl (start crawl)
↓
Wait Node (60 seconds)
↓
HTTP Request → GET from Firecrawl (fetch results)
↓
Split Out (10 pages → process one at a time)
↓
Message a Model (GPT-4o-mini analyses each page)
↓
Code Node (parse JSON output into clean rows)
↓
Google Sheets (append each event as a new row)

```

## 5. The System Prompt - Where Product Thinking Lives

The system prompt is the most important part of the entire agent. It encodes the business logic, the rules about what to track, how to name events, what properties to include, and how to prioritise. This is pure PM thinking translated into instructions for an AI.

### The Prompt

```
You are a Mixpanel analytics expert working on a  
financial learning platform called NetworkFP.
```

```
RULES:
```

1. Always include `page_viewed` as the first event
2. Identify every action a user can take on this page
3. Use `snake_case` for all event names
4. Keep property names short and clear
5. Only suggest events useful for understanding behaviour
6. Add `page_url` field to every event

```
For each event output:
```

- `event_name`
- `page_url`
- `properties` (comma separated)
- `priority`: High / Medium / Low
- `reason`: one line why this matters

```
Output as JSON array only. No explanation.
```

### Why This Prompt Is A Product Decision

Every rule in the prompt reflects a product judgment:

- `page_viewed` always first because without basic page tracking, nothing else makes sense
- `snake_case` enforced because Mixpanel convention requires it and inconsistent naming breaks dashboards
- Properties kept short because verbose property names create messy reports
- Priority field because developers can't implement everything at once and need to know what matters most
- Reason field because without justification, developers skip events they don't understand

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## 6. The Output

### What The Google Sheet Looks Like

Every morning, new rows are added to the sheet automatically. Here is a sample of what the output looks like:

|    | A          | B                   | C                   | D                                              | E        | F                                                               |
|----|------------|---------------------|---------------------|------------------------------------------------|----------|-----------------------------------------------------------------|
| 1  | Date       | Page URL            | Event Name          | Properties                                     | Priority | Reason                                                          |
| 2  | 2026-04-18 | learn.networkfp.com | page_viewed         | page_url, page_title, user_id, timestamp       | High     | Tracks every page visit, foundation of all analytics            |
| 3  | 2026-04-18 | learn.networkfp.com | lesson_completed    | lesson_id, course_name, user_id, timestamp     | High     | Indicates user progress and engagement in learning              |
| 4  | 2026-04-18 | learn.networkfp.com | quiz_attempted      | quiz_id, course_name, user_id, timestamp       | High     | Measures interaction with quizzes and understanding of materi   |
| 5  | 2026-04-18 | learn.networkfp.com | resource_downloaded | resource_id, resource_type, user_id, timestamp | Medium   | Tracks valuable content engagement and resource utility         |
| 6  | 2026-04-18 | learn.networkfp.com | rating_submitted    | lesson_id, rating_value, user_id, timestamp    | Medium   | Gathers feedback to assess content quality and user satisfactio |
| 7  | 2026-04-18 | learn.networkfp.com | comment_posted      | lesson_id, comment_text, user_id, timestamp    | Medium   | Captures user engagement and interaction with community         |
| 8  | 2026-04-18 | learn.networkfp.com | bookmark_added      | lesson_id, user_id, timestamp                  | Low      | Indicates user interest in revisiting specific content          |
| 9  | 2026-04-18 | learn.networkfp.com | page_viewed         | page_url, page_title, user_id, timestamp       | High     | Tracks every page visit, foundation of all analytics            |
| 10 | 2026-04-18 | learn.networkfp.com | course_enrolled     | course_id, user_id, timestamp                  | High     | Indicates user interest in specific courses                     |
| 11 | 2026-04-18 | learn.networkfp.com | quiz_started        | quiz_id, course_id, user_id, timestamp         | High     | Measures user engagement with quizzes                           |
| 12 | 2026-04-18 | learn.networkfp.com | quiz_completed      | quiz_id, course_id, score, user_id, timestamp  | High     | Tracks completion and performance in quizzes                    |
| 13 | 2026-04-18 | learn.networkfp.com | article_shared      | article_id, user_id, share_platform, timestamp | Medium   | Measures content reach and user advocacy                        |
| 14 | 2026-04-18 | learn.networkfp.com | resource_downloaded | resource_id, course_id, user_id, timestamp     | Medium   | Indicates interest in downloadable materials                    |
| 15 | 2026-04-18 | learn.networkfp.com | feedback_submitted  | feedback_type, user_id, timestamp              | Medium   | Captures user input for platform improvement                    |
| 16 | 2026-04-18 | learn.networkfp.com | video_completed     | video_id, course_name, user_id, timestamp      | Medium   | Measures full engagement with video content                     |

## How The Developer Uses This

The developer receives this sheet and implements each event directly into the codebase.

## 7. Key Product Decisions Made

Several important product decisions were made while designing this agent:

### Decision 1: Auto-discover pages, don't define manually

Initially the workflow manually listed 10 pages. This was identified as a flaw - manual input doesn't scale and someone has to maintain the list. Firecrawl's auto-crawl mode was used instead, which discovers all pages automatically from the homepage - just like Google's crawler does.

### Decision 2: Embed URL in the AI prompt, not the pipeline

Getting the page URL to survive through the n8n pipeline proved technically complex - n8n's OpenAI node drops input fields from previous nodes. The solution was to embed the URL instruction inside the prompt itself so the AI carries it within every event object. This is a design pattern worth remembering: when data gets lost in a pipeline, make the AI carry it.

### Decision 3: Output to Google Sheets, not directly to Mixpanel

Events were not auto-implemented into Mixpanel. A human review step was kept deliberately - the sheet goes to a developer who validates and implements. This is the right call because AI can miss business context, and incorrect event implementation is worse than no implementation.

### Decision 4: Limit to 10 pages per daily run

Firecrawl's free tier allows 500 pages/month. Crawling 10 pages/day = 300 pages/month, well within the limit. This also keeps OpenAI costs negligible at under ₹1/month.

## 8. Impact & Results

- Reduced analytics instrumentation planning from 20-30 days to fully automated overnight
  - Zero manual effort after initial setup - runs every morning without anyone touching it
  - Total infrastructure cost: ₹0/month
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